

Urban Heat Island & Vulnerability Analysis — Phoenix Metropolitan Area, Arizona

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1. Introduction

The Urban Heat Island (UHI) effect, characterized by significantly elevated surface temperatures in urbanized areas relative to surrounding rural landscapes, represents a pressing environmental and public health crisis in rapidly expanding desert metropolises. In Phoenix, Arizona — one of the fastest-growing cities in the United States — this phenomenon is exacerbated by dense concentrations of impervious surfaces that absorb and re-radiate solar energy at rates far exceeding natural desert substrates. However, heat exposure is not uniformly distributed across the urban landscape, nor are its consequences felt equally among residents.

Building upon environmental justice frameworks, this study investigates the spatial intersection of physical heat distribution and socioeconomic vulnerability across Maricopa County, Arizona. Land Surface Temperature (LST), vegetation cover, and impervious surface density were derived from Landsat 9 satellite imagery and integrated with U.S. Census demographic data to construct a composite spatial vulnerability index. A Getis-Ord G_i^* hot spot analysis was subsequently applied to identify statistically significant clusters of high heat vulnerability. The results provide spatially explicit, actionable insights for urban heat mitigation and demonstrate the capacity of NASA Earth observation data to advance environmental equity.

2. Study Area

The study area encompasses Maricopa County, Arizona, with analytical focus on the Phoenix Metropolitan Area. Located within the northern Sonoran Desert, the region experiences extreme summer temperatures that frequently exceed 40°C, with surface temperatures over impervious materials reaching significantly higher values. The landscape presents a dramatic contrast between a densely developed, low-albedo urban core and a surrounding desert environment punctuated by irrigated parks, golf courses, agricultural zones, and riparian corridors along the Salt River.

This geographic context makes Maricopa County an ideal case study for UHI analysis. The stark thermal contrast between natural and built surfaces is clearly resolvable at Landsat's 30-meter spatial resolution, and the county's rapid sprawl history introduces significant socioeconomic heterogeneity across its 1,009 census tracts — a critical condition for meaningful environmental justice analysis.

3. Data & Methods

3.1 Data Acquisition

Six core datasets were acquired and processed to establish physical and demographic baselines for the year 2023:

Landsat 9 OLI/TIRS Collection 2 Level-2 (USGS EarthExplorer): A single scene acquired on August 25, 2023 (Scene ID: LC09_L2SP_037037_20230825_20230827_02_T1) was selected from three candidate scenes on the basis of minimal cloud cover over the Phoenix urban core. Monsoon season convective cloud development was present in the southern portion of the scene but did not affect the primary study area. Band 10 (ST_B10) was utilized for LST derivation; Bands 4 (Red) and 5 (NIR) were used for NDVI calculation.

NLCD 2023 Fractional Impervious Surface — CONUS (MRLC): Annual NLCD Collection 1 product providing sub-pixel impervious surface density at 30-meter resolution. Values range from 0–100%, representing the percentage of each pixel covered by impervious material. The 2023 product was selected to match the Landsat scene acquisition year.

ACS 2023 5-Year Estimates (U.S. Census Bureau, data.census.gov): Three demographic tables were extracted at the census tract level for Maricopa County: Table S1901 (Median Household Income), Table S0101 (Age and Sex, used to derive the population aged 65 and over), and Table B02001 (Race, used to derive the non-white population percentage).

TIGER/Line Census Tracts 2023 (U.S. Census Bureau): Spatial tract boundaries for Arizona, filtered to Maricopa County (COUNTYFP = '013'), used as the zone dataset for zonal statistics and as the spatial join target for all demographic tables.

3.2 Data Processing & Spatial Alignment

All datasets were harmonized into a common spatial reference system: NAD83 UTM Zone 12N (EPSG: 26912). Landsat bands were reprojected using bilinear resampling; the NLCD raster was reprojected using nearest neighbor resampling to preserve categorical integrity. All rasters were clipped to the Maricopa County tract boundary using the Extract by Mask tool in ArcGIS Pro.

3.3 Physical Indicator Derivation

Land Surface Temperature (LST): Derived from the Landsat 9 Level-2 ST_B10 band using the standard USGS scale factor conversion:

$$\text{LST (}^{\circ}\text{C)} = (\text{ST_B10} \times 0.00341802 + 149.0) - 273.15$$

Normalized Difference Vegetation Index (NDVI): Calculated from surface reflectance Bands 4 and 5:

$$\text{NDVI} = (\text{Band 5} - \text{Band 4}) / (\text{Band 5} + \text{Band 4})$$

Both calculations were executed using the ArcGIS Pro Raster Calculator.

3.4 Vulnerability Index Construction

Zonal statistics (mean) were calculated for LST, NDVI, and fractional impervious surface across all 1,009 census tracts using Maricopa_Tracts_Reproject.shp as the zone dataset. ACS demographic tables were cleaned prior to joining: the Census label row was removed, GEO_ID prefix values (1400000US) were stripped and padded to 11-digit FIPS format, and non-numeric income entries were standardized. All tables were permanently joined to the tract shapefile via GEOID, validated at 1,009 matching records.

Two derived demographic fields were calculated:

$$\text{PCT_65PLUS (\%)} = (\text{POP_65PLUS} / \text{TOTAL_POP}) \times 100$$

$$\text{PCT_NONWHT (\%)} = ((\text{TOTAL_POP} - \text{WHITE_ALONE}) / \text{TOTAL_POP}) \times 100$$

All six vulnerability variables were normalized to a 0–1 scale using min-max normalization.

Income and NDVI were inverted so that lower values correspond to higher vulnerability scores:

$$\text{NORM_LST} = (\text{MEAN_LST} - 50.537) / (65.317 - 50.537)$$

$$\text{NORM_NDVI} = 1 - ((\text{MEAN_NDVI} - 0.040) / (0.234 - 0.040))$$

$$\text{NORM_IMP} = (\text{MEAN_IMP} - 0.000) / (82.547 - 0.000)$$

$$\text{NORM_INCOME} = 1 - ((\text{MED_INC} - 0) / (250000 - 0))$$

$$\text{NORM_65PLUS} = \text{PCT_65PLUS} / 100$$

$$\text{NORM_NONWHT} = \text{PCT_NONWHT} / 100$$

The Composite Vulnerability Index (VULN_INDEX) was calculated using the following weighted combination:

$$\text{VULN_INDEX} = (\text{NORM_LST} \times 0.30) + (\text{NORM_IMP} \times 0.20) + (\text{NORM_INCOME} \times 0.20) + (\text{NORM_NDVI} \times 0.15) + (\text{NORM_65PLUS} \times 0.10) + (\text{NORM_NONWHT} \times 0.05)$$

Weight rationale: LST (0.30) was assigned the highest weight as the direct measure of heat exposure. Impervious surface (0.20) and income (0.20) were weighted equally as the primary physical UHI driver and strongest socioeconomic vulnerability predictor, respectively. NDVI (0.15) captures the cooling effect of vegetation. Age 65+ (0.10) reflects physiological vulnerability to heat stress. Race and ethnicity (0.05) provides environmental justice framing while avoiding overweighting a single demographic indicator.

3.5 Spatial Statistics

A Getis-Ord Gi* Hot Spot Analysis was executed on the VULN_INDEX field using a Fixed Distance Band conceptualization and Euclidean distance method. ArcGIS Pro calculated a default neighborhood search threshold of 45,293.59 meters based on the spatial distribution of tract centroids.

4. Results

4.1 Land Surface Temperature

Pixel-level LST values across the study area ranged from -21.5°C to 77.9°C. The minimum values correspond to water bodies northeast of Phoenix (Lake Pleasant and Roosevelt Lake), where the thermal retrieval algorithm produces known sub-zero artifacts over deep water during atmospheric correction. Maximum values correspond to bare desert and dense impervious urban surfaces exposed to direct August solar radiation. The UHI signal is clearly visible in the spatial distribution of LST: the Phoenix urban core exhibits dramatically elevated surface temperatures relative to surrounding desert, with the Salt River corridor and irrigated vegetated areas appearing as pronounced thermal cool zones.

4.2 NDVI

NDVI values ranged from -0.400 to 0.645 across the study area. Negative values correspond to water bodies. Low values (0.0–0.1) dominate the urban core and bare desert surfaces, consistent with minimal photosynthetically active vegetation. Higher values (0.3–0.6) are concentrated along the Salt River corridor and within irrigated suburban areas — golf courses, parks, and landscaped residential neighborhoods. A strong inverse spatial relationship between NDVI and LST is visually and analytically confirmed: areas of highest vegetation density correspond directly to the lowest surface temperatures.

4.3 Fractional Impervious Surface

Impervious surface values ranged from 0–100% per 30-meter pixel. The Phoenix urban core exhibits the highest impervious density, with large contiguous areas exceeding 71% impervious cover. Suburban development rings radiate outward at intermediate impervious densities (30–70%), while the surrounding desert and reservation lands record near-zero values. The road network radiating outward from the urban center is clearly resolved as a fractional impervious signal through otherwise low-density areas.

4.4 Composite Vulnerability Index

VULN_INDEX scores across the 1,009 Maricopa County census tracts ranged from 0.178 to 0.782, with a mean of 0.558 and a standard deviation of 0.095. The distribution is approximately normal with a slight right skew. No null values were present across all 1,009 records. The spatial pattern of the vulnerability index strongly mirrors the LST and impervious surface distributions, with the highest scores concentrated in the Phoenix urban core and transitioning to lower scores in suburban and rural tracts.

4.5 Hot Spot Analysis

The Getis-Ord Gi* analysis confirmed strong, statistically significant spatial clustering of heat vulnerability across Maricopa County. A dense, contiguous cluster of 99% confidence hot spots is concentrated within the Phoenix urban core, corresponding to tracts characterized by high LST, extensive impervious cover, low median income, and minimal vegetation. Cold spots at

99% confidence are located in large rural tracts to the west and northeast, including the Tonto National Forest area, driven by near-zero impervious footprints and higher relative incomes. Tracts within the Gila River Indian Reservation returned as not statistically significant, a spatial pattern discussed further in Section 5.

5. Discussion

The results of this analysis reveal a clear and statistically significant pattern of environmental injustice within the Phoenix Metropolitan Area. The spatial convergence of high surface temperatures, extensive impervious cover, low vegetation density, and lower median household income in the urban core identifies a population that faces compounded heat vulnerability with limited adaptive capacity. Residents of these tracts are simultaneously exposed to the greatest thermal stress and least equipped, financially and infrastructurally, to mitigate it through measures such as air conditioning, green space access, or relocation.

The inverse relationship between NDVI and LST confirms that urban greening functions as an effective thermal mitigation mechanism. However, the current distribution of vegetation strongly favors affluent suburban areas, golf courses, landscaped estates, and irrigated parks, rather than the dense, lower-income urban tracts where cooling is most critically needed. This inequitable distribution of vegetative cooling capacity reinforces existing socioeconomic disparities and underscores the need for targeted urban greening interventions in vulnerable communities.

The Gila River Indian Reservation tracts merit specific discussion. These tracts returned as not statistically significant in the hot spot analysis despite their proximity to the urban core. This likely reflects a combination of factors: distinct land management practices, lower housing and impervious surface densities, and spatial configurations that break the contiguous high-impervious, low-income pattern driving the urban hot spot cluster. This finding illustrates the importance of contextualizing spatial statistics within the broader cultural and land-use histories of study area communities rather than interpreting Gi* results in isolation.

More broadly, this study demonstrates the power of NASA Earth observation assets, specifically Landsat 9, to generate spatially explicit, equity-relevant urban climate information at the census tract scale. The integration of remotely sensed physical indicators with socioeconomic demographic data provides a replicable methodological framework applicable to any rapidly urbanizing region facing extreme heat risk.

6. Limitations & Future Work

Several technical and methodological limitations should be acknowledged:

LST Water Body Artifacts: The minimum calculated LST of -21.5°C represents a known artifact over deep water bodies northeast of Phoenix, where the Landsat Level-2 atmospheric correction algorithm overcompensates during moisture adjustment. While these values are smoothed during zonal averaging at the tract level and do not materially affect the vulnerability analysis, future iterations should apply a pre-calculation null mask over water bodies to eliminate these artifacts entirely.

Single-Date Snapshot: This analysis relies on a single Landsat scene from August 25, 2023. While this date captures peak summer thermal conditions, it cannot account for temporal variability in LST or vegetation cover across the season. A multi-temporal analysis using seasonal composites would produce a more robust characterization of chronic heat exposure.

Modifiable Areal Unit Problem (MAUP): The choropleth vulnerability map is subject to the Modifiable Areal Unit Problem, whereby large-area rural tracts in western Maricopa County visually dominate the map despite low population densities. Census tract boundaries are administrative units that do not correspond to thermal or demographic homogeneity, and aggregation to the tract level necessarily suppresses within-tract variability.

Sensor Resolution: Landsat's 30-meter spatial resolution, while appropriate for county-scale analysis, limits the detection of fine-scale urban thermal features such as individual street trees, building shadows, or small green spaces. Higher-resolution thermal sensors such as ECOSTRESS (70-meter) or future TIRS-3 products would enable more granular vulnerability mapping.

Future Directions: Incorporating nighttime LST data from ECOSTRESS or NOAA-20 VIIRS would substantially improve the health relevance of the index, as the most dangerous aspect of UHI exposure is the failure of urban surfaces to cool overnight — a condition that prevents human physiological recovery from daytime heat stress. Future work could also incorporate cooling center accessibility, tree canopy equity indices, and multi-year LST trend analysis to produce a more comprehensive heat vulnerability assessment.

7. References

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